



Weekly Macro Outlook

Model-Based Forecasts

US CPI | US PPI | US Retail Sales | UK Labour Market
MIDAS Features, Lagged PCA, Ridge/ElasticNet/XGBoost

Week of May 12–15, 2026

Methodology

Framework

For each target variable, we run a systematic model search: 5 models $\times$ 4 feature sets = 20 combinations, evaluated on a hard train/test split (train: pre-2023, test: 2023–2025). COVID months (2020-02 to 2020-08) excluded from training. Only models passing the overfitting gate (test RMSE $< 1.5 \times$ CV RMSE) qualify for inference.

Feature Construction

All features are strictly **lagged** (no contemporaneous information). High-frequency predictors (daily: oil, VIX, yields, FX; weekly: claims) are aggregated to monthly via MIDAS Beta polynomial weights ($\theta_1 = 1, \theta_2 = 5, K=22$ for daily, $K=4$ for weekly). Same-frequency regressors enter with 1-month lag.

Multicollinearity Treatment

1. **StandardScaler**: all features standardised to zero mean, unit variance
2. **VIF elimination**: iteratively drop highest-VIF feature until all VIF < 10
3. **PCA**: fit on train-only (ex-COVID), extract 3–5 orthogonal factors, **lag by 1 month**

Principal Component Eigenstructure

PCA is fitted on the standardised lagged predictor universe using the training sample excluding COVID. Each principal component is an eigenvector of the predictors' covariance matrix, and its eigenvalue measures the variance explained by that component. The retained PCs therefore represent orthogonal macro channels ordered by their eigenvalues.

- **PC1**: the largest eigenvalue, capturing the broad monetary policy / yield regime channel.
- **PC2**: the second-largest eigenvalue, capturing curve slope and demand-pressure dynamics.
- **PC3**: the third component, capturing energy / supply-side pressure and real activity momentum.
- **PC4**: a confidence / wealth/income channel with lower but meaningful variance.
- **PC5**: a financial stress / risk-premium channel, the smallest retained eigenvalue among the five.

The regression coefficients on PC1–PC5 are therefore best read as sensitivities to these orthogonal eigenvector channels rather than individual raw predictors.

Model Specifications

- **OLS**: unpenalised linear regression (baseline)
- **Ridge**: L2 penalty, λ by LOO-CV — shrinks correlated coefficients uniformly
- **ElasticNet**: L1+L2 mix, automatic feature selection via sparsity
- **XGBoost**: gradient-boosted trees (depth 2–4, L2=2–5, lr=0.03–0.05)

Coefficients shown in this report are taken directly from the backend results file and represent standardised linear weights for each selected model. They are used to identify the dominant macro channels and to explain the economic drivers behind each forecast.

Probabilistic Output

Point forecast: best model refitted on full ex-COVID sample. **Confidence intervals**: Quantile Regression at $\tau \in \{0.10, 0.25, 0.50, 0.75, 0.90\}$. **Beat/miss**: Ensemble of Ordered Logit (25%) + XGBoost Classifier (50%) + Quantile Regression (25%).

US CPI YoY — Tue May 12 (8:30 ET)

Best Model: ElasticNet + Lagged PCA(5)

$$\widehat{\text{CPI}}_t^{\text{YoY}} = \alpha + \sum_{j=1}^5 \beta_j \cdot F_{j,t-1} + \gamma \cdot \text{CPI}_{t-1}^{\text{YoY}} + \varepsilon_t$$

Test $R^2 = 0.872$ (20 models tested, 19/20 passed gate). Classification accuracy: 60%. The PC channels are orthogonal eigenvectors and their eigenvalues determine the variance explained by each component.

Estimated Coefficients (standardised)

Variable	Coef.	Interpretation
CPI_YOY_lag1	+0.993	Near unit-root persistence — inflation is highly autocorrelated
PC5_lag1	+0.151	Financial conditions factor (VIX/credit-spread loaded)
PC3_lag1	+0.097	Energy-price factor (oil-weighted component)
PC2_lag1	−0.100	Yield curve slope — tighter policy dampens future inflation
PC1_lag1	−0.026	Rates level — higher rates reduce demand-pull inflation
PC4_lag1	−0.007	Near-zero: minimal incremental contribution

The dominant predictor is lagged CPI itself ($\beta = 0.993$), reflecting strong inflation persistence from wage-price indexation and administered price contracts. The PCA factors add marginal information: financial stress (PC5) pushes inflation up, while tighter monetary policy (PC2, slope) pushes it down.

Forecast

Previous	Consensus	Point	Median	80% CI	P(miss)	P(beat)
3.3%	3.75%	2.58%	2.58%	[2.19, 2.82]	28.8%	12.2%

Interpretation: The model forecasts 2.58% YoY, well below the 3.75% consensus. The 80% CI [2.19, 2.82] does not contain the consensus. $P(\text{miss})=59\%$ dominates. The model relies on lagged CPI (3.3% in March) and sees the energy pass-through as already partially embedded in the base, while the yield curve tightening signal (PC2) dampens forward inflation. *Caveat:* the model uses data through March; it cannot capture intra-April energy price developments that the consensus may incorporate.

US PPI MoM — Wed May 13 (8:30 ET)

Best Model: ElasticNet + VIF-Cleaned Features

Test $R^2 = -0.154$ (all 20 models passed gate). ElasticNet selected with strong L1 sparsity — only a few variables retain material weight:

Variable	Coef.	Interpretation
PPI_MOM_lag1	+0.433	Strong positive momentum from last month
PPI_MOM_lag2	+0.084	Additional persistence over a two-month horizon
VIX_midass	-0.040	Higher uncertainty → demand destruction → lower producer prices
<i>All others</i>	0.000	Zeroed by L1 penalty (oil, credit spread, EUR/USD, level, CPI not selected)

Prev	Cons	Point	Median	80% CI	P(miss)	P(beat)
+0.5%	+0.5%	+0.96%	+1.56%	[+0.11, +2.64]	10.6%	34.2%

Interpretation: The model forecasts above consensus (0.96% vs 0.5%), driven by the lagged momentum from March's +0.5%. P(beat)=34% outweighs P(miss)=11%. Oil was zeroed by ElasticNet, suggesting that the oil-to-PPI pass-through lag exceeds 1 month in the current regime.

US Retail Sales MoM — Thu May 14 (8:30 ET)

Best Model: Ridge + Lagged PCA(5)

Test $R^2 = +0.049$. Modest but positive — consumption is driven by a mix of wealth, confidence, and income effects that PCA captures as a composite. The retained PCs are eigenvectors of the predictor covariance matrix, ordered by the variance each eigenvalue explains.

Variable	Coef.	Interpretation
US_RETAIL_MOM_lag1	-0.141	Mean reversion in retail sales after the February/March spike
PC2_lag1	-0.115	Tighter macro/financial conditions damp demand
PC4_lag1	+0.075	Positive consumer confidence/wealth signal supporting spending
PC5_lag1	+0.023	Risk-appetite or global demand lift
PC3_lag1	-0.019	Small drag from residual energy/confidence factor
PC1_lag1	+0.001	Near-zero: policy-rate / term structure effect is negligible

Prev	Cons	Point	Median	80% CI	P(miss)	P(beat)
+1.7%	+0.3%	+0.37%	+0.46%	[-0.67, +1.09]	26.5%	14.3%

Interpretation: The model sees a sharp normalisation from March's +1.7% to +0.37%, broadly in line with consensus. P(inline)=59% dominates. The wide 80% CI reflects the high noise in monthly retail data. The mean-reversion from March's spike is the dominant signal.

UK Labour Market — Wed May 13 (07:00 BST)

Best Model: Ridge + Lagged PCA(5)

Test $R^2 = 0.914$. The highest R^2 across all 4 targets. UK unemployment is highly persistent and well-captured by the yield curve regime (PC1/PC2 encode BoE expectations). These PCs are eigenvectors of the standardized predictor covariance matrix, with eigenvalues ranking their variance contribution.

Variable	Coef.	Interpretation
UK_UNEMP_lag1	+1.244	Very strong persistence in the unemployment rate
PC1_lag1	−0.207	Strong macro/regime factor: tighter BoE expectations reduce unemployment slightly
PC2_lag1	−0.069	Additional policy/labour slack signal that tempers jobless growth
PC4_lag1	+0.083	Financial stress or demand weakness raising unemployment
PC5_lag1	+0.104	Global risk/real-economy pressure lifting joblessness
PC3_lag1	−0.015	Very small negative contribution from residual cyclical factor

Prev	Cons	Point	Median	80% CI	P(miss)	P(beat)
4.9%	4.9%	5.13%	5.09%	[4.87, 5.28]	0.4%	18.3%

Interpretation: The model forecasts 5.13%, above the 4.9% consensus. $P(\text{inline})=81\%$ dominates but $P(\text{beat})=18\%$ is non-trivial — the model detects upward drift consistent with the rising trend (4.4% → 4.9% over the past year, vacancies at 5-year low). This would signal further labour market deterioration ahead of the June 18 BoE MPC decision.

Cross-Variable Coherence

The four forecasts paint a coherent macro picture:

- **CPI below consensus + PPI above consensus** (→ margin squeeze): CPI at 2.58% vs PPI at 0.96% MoM suggests producer costs are rising faster than consumer prices — a margin squeeze for corporates. This is consistent with the energy cost pass-through still working through the supply chain upstream (PPI) before reaching the consumer (CPI).
- **Retail inline** (→ consumer absorbing shocks): The 0.37% retail forecast is close to consensus, suggesting the consumer is neither collapsing (which would be deflationary) nor overheating (which would add demand-pull pressure). This validates the “soft landing” narrative.
- **UK unemployment drifting higher** (→ BoE dilemma deepens): At 5.13%, UK unemployment is rising while CPI remains elevated (3.3% YoY in March). This is the textbook stagflationary signal: the BoE cannot cut to support employment without risking re-igniting inflation, and cannot hike without deepening the labour market deterioration.
- **Global spillover:** A below-consensus US CPI would be *dovish* for the Fed (supporting rate cuts), which would compress US yields and — via our VAR spillover analysis — reduce pressure on gilt yields (29% of UK Level FEVD from US). This would partially offset the hawkish domestic signal from rising UK unemployment.

Data: FRED (34 series: CPI, PPI, retail sales, UK unemployment, oil, VIX, yields, FX, claims, etc.), NS factors (5 countries, daily). 20 model combinations per target, COVID-excluded training. This document does not constitute investment advice.